Normalization Assignment

Assignment: HR Database Normalization

Scenario

A company stores employee HR data in one spreadsheet.

emp_id	emp_name	department	manager	skills	certifications	project	project_location
E01	Aung Aung	IT	U Hla	Python, SQL	AWS, Azure	HRIS	Yangon
E02	Su Su	Finance	Daw May	Excel, Power BI	CPA	Payroll	Mandalay
E03	Kyaw Kyaw	IT	U Hla	Java, Python	AWS	HRIS	Yangon

Part A: Questions

Q1. What normalization problem exists in the original HR table?

Answer:

The table contains repeating groups and multiple values in one column, such as skills and certifications. It also mixes employee, department, manager, project and location data in one table. This causes redundancy and update problems.

Q2. Convert the HR table into 1NF.

Answer:

In 1NF, each column must contain atomic values.

emp_id	emp_name	department	manager	skills	certifications	project	project_location
E01	Aung Aung	IT	U Hla	Python	AWS	HRIS	Yangon
E01	Aung Aung	IT	U Hla	SQL	Azure	HRIS	Yangon
E02	Su Su	Finance	Daw May	Excel	CPA	Payroll	Mandalay
E02	Su Su	Finance	Daw May	Power BI	CPA	Payroll	Mandalay
E03	Kyaw Kyaw	IT	U Hla	Java	AWS	HRIS	Yangon
E03	Kyaw Kyaw	IT	U Hla	Python	AWS	HRIS	Yangon

Q3. What is the main issue after converting to 1NF?

Answer:

Employee, department, manager, and project data are repeated.

Q4. Convert the 1NF table into 2NF.

Answer:

In 2NF, remove partial dependency. Separate data that depends only on part of a composite key.

Employees

emp_id	emp_name	dept_id
E01	Aung Aung	D01
E02	Su Su	D02
E03	Kyaw Kyaw	D01

Departments

dept_id	department	manager
D01	IT	U Hla
D02	Finance	Daw May

Employee_Skills

emp_id	skill
E01	Python
E01	SQL
E02	Excel
E02	Power BI
E03	Java
E03	Python

Employee_Certifications

emp_id	certifications
E01	AWS
E01	Azure
E02	CPA
E03	AWS

Employee_Projects

emp_id	project	project_location
E01	HRIS	Yangon
E02	Payroll	Mandalay
E03	HRIS	Yangon

Q5. What issue still exists after 2NF?

Answer:
project_location depends on project, not directly on emp_id. This is a transitive dependency.

Q6. Convert the 2NF design into 3NF.

Answer:
In 3NF, remove transitive dependency.

Projects

project_id	project	project_location
P01	HRIS	Yangon
P02	Payroll	Mandalay

Employee_Projects

emp_id	project_id
E01	P01
E02	P02
E03	P01

Q7. Is the HR database now in BCNF?

Answer:
Yes, if every determinant is a candidate key.
Example:

- emp_id → emp_name, dept_id
- dept_id → department, manager
- project_id → project_name, project_location

Q8. What 4NF issue can exist in HR data?

Answer:
An employee can have multiple independent skills and multiple independent certifications. These should not be stores together in one table because they create unnecessary combinations.

Wrong design:

emp_id	skill	certification
E01	Python	AWS
E01	Python	Azure
E01	SQL	AWS
E01	SQL	Azure

Correct 4NF design:

Employee_Skills

emp_id	skill
E01	Python
E01	SQL

Employee_Certifications

emp_id	certification
E01	AWS
E01	Azure

Q9. What is a possible 5NF issue in HR data?

Answer:
A complex relationship may exist among employees, projects, and roles.

Wrong design:

emp_id	project_id	role
E01	P01	Developer
E02	P02	Tester
E03	P01	Accountant

If employee-project, employee-role, and project-role relationships are independent, they can be separated.

Employee_Project

emp_id	project_id
E01	P01
E02	P02
E03	P01

Employee_Role

emp_id	role
E01	Developer
E02	Tester
E03	Accountant

Project_Role

project_id	role
P01	Developer
P02	Tester
P03	Accountant

Q10. What is the final normalized HR database design?

Answer:

Final tables:

1. Employees
2. Departments
3. Projects
4. Employee_Skills
5. Employee_Certifications
6. Employee_Projects
7. Optional: Roles, Employee_Roles, Project_Roles